

A Layer Jamming Soft Glove for Hand Tremor Suppression

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Abstract—Tremors are a common movement disorder that affects a person's life adversely. With various drawbacks of current treatment methods, there is a need for a mechanical solution. The authors present a soft orthosis based on layer jamming for the suppression of hand tremors. A vacuum supplied to the layer jamming elements which contain a stack of layers attached to the glove leads to increased stiffness in the glove, suppressing the tremor. The behavior of the tremor in a cohort of patients in Sri Lanka was studied and showed that the tremor's mean frequency was 5.05 ± 2.03 Hz. An existing analytical model was modified and verified experimentally to select the layer jamming element. An element with sandpaper (320 grade) and tracing paper was chosen based on the high stiffness provided at a lower weight (total weight of the glove = 30g). Experimental results show minimal variation in stiffness for vacuum pressures between 10-50 kPa (abs.). The performance evaluation on a test rig was used to verify the efficacy of the glove and showed optimal placement on the palmar side with a mean tremor amplitude reduction of 78.32%. Clinical trials conducted on 11 recruited tremor patients showed a mean frequency power reduction of $41.74 \pm 12.11\%$, $41.99 \pm 14.82\%$, and $24.71 \pm 12.18\%$ in the index and middle fingers and in grasping, respectively with a maximum power reduction of 59.15%. The soft, low weight orthosis shows a high tremor suppression in both the test rig and clinical experiments.

Index Terms—Tremor suppression, hand tremor, soft robotics, layer jamming, orthotics.

I. INTRODUCTION

TREMORS are a common motion disorder associated with most neuromuscular diseases [1]. A tremor is an involuntary, and occasionally rhythmic, contraction of muscles of any part of the body [2]. Limbs, head, torso, and vocal cords

are common sites where tremors are observed. Although most tremors lack a definite cause, common neurodegenerative diseases such as Parkinson's disease, neurological disorders such as brain injuries, strokes and multiple sclerosis, chemical abuse or medicinal side effects, are known to cause tremors [2].

Tremors are not generally life threatening but despite being generally benign, they can affect a patient's daily lifestyle in numerous ways [3]. Tremors, especially those occurring in the hands, can hinder a patient's ability to conduct activities of daily living (ADL). Additionally, they can affect a patient's psychology negatively [3].

Tremors are classified based on several categories according to their behavior or associated condition [4]. Generally, these are classified as resting and action, based on the working condition of the patient [1]. Action tremors affect ADL. Although a high number of tremors are noted based on appearance, essential tremors and Parkinsonian tremors are widespread and well known [2]. According to current studies, essential tremors affect up to 5% of the population above the age of 65 and Parkinsonian tremors affect up to 1% above the age of 65 [1]. Mechanically tremors are classified based on their frequency. Contemporary studies show that the tremor frequency varies between 4-18 Hz generally [2].

As mentioned previously, tremors can affect a patient's daily life negatively. Therefore, a solution could be presented to reduce tremors. Medically, tremors are reduced by two main methods. They are through medication or surgery [5], [6]. Medication, commonly beta-blockers and medication for Parkinson's disease, has side effects associated with them including reduced blood pressure and heart rate, fatigue, depression, and behavioral changes [7]. Surgeries and invasive interventions such as deep brain stimulation (DBS) and thalamotomy can be performed to suppress the tremors [6], [8]. DBS introduces a permanent electrode into the brain to stimulate while thalamotomy destroys part of the thalamus. These surgeries may lead to permanent brain damage or comas and in extreme cases, even death. Therefore, the need for a mechanical solution to suppress tremors can be presented [9], [10].

Several hand tremor suppression devices can be found in the literature. Each of these devices have different features in tremor suppression. Upon, reviewing currently available literature, both exoskeletal devices and soft orthoses can be seen.

These devices can be broadly categorized based on the point or method of tremor suppression. Most tremor suppression devices focus on suppressing tremors either in the

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forearm or wrist, while a lesser number of such orthoses are designed for the suppression of tremors in the phalanges of the hand [11]. Considering that the available research on hand tremors shows that the tremor frequency increases in distal joints, there is a higher requirement in suppressing tremors in these distal joints compared to proximal joints [12]. Many contemporary tremor suppression devices are only capable of suppressing tremors around the wrist or forearm due to the high requirement of such joints for many hand movements. But tremors also appear on finger joints which are required in high dexterity hand movements such as the power pinch. The tremors propagate through mechanically unrestricted joints towards restricted joints [12]. Therefore, applying restriction to the tremor motion in finger joints is also necessary to suppress the hand tremor completely from the hand.

These devices suppress tremors in several methods. Exoskeletal devices are more commonly seen. These devices are actuated using DC motors to control the tremor actively [13], [14]. These exoskeletal devices were made with bulky and rigid structures and components which added high weight to the wearer, ranging from 850 g to 1700 g. Orthoses with active magneto-rheological dampers are also available in research [15]. Glove based wearable designs are driven either by tendons or pneumatic actuator muscle (PAM) [16]–[19]. Zhou *et al.* proposed a tendon driven wearable tremor suppression glove that weighs 300 g. Such mechanisms are carrying the risk of injuries to the user primarily caused by component malfunctioning [11]. These designs suppress the tremor actively with the use of feedback from attached sensors. On the other hand, passive or semi-active devices used in the suppression of tremors are mainly based on the absorption of energy from the tremor leading to the damping especially with the use of pneumatic based designs. Moreover, Fromme *et al.* proposed a pneumatically driven passive orthosis, suppressing tremors in the wrist joint [18]. Even though this method is passive and light weight, the air structure of this orthosis becomes bulkier during inflation.

The use of active wearable devices usually leads to a higher peripheral loading on limbs. Such peripheral loading of limbs, especially those weakened due to neuromuscular diseases, leads to fatigue for the patient wearing the device. Hence these are not most applicable. A majority of such devices are neither ergonomically viable nor have an adequate aesthetic appearance. Therefore, patients might avoid such solutions.

Thus, when considering contemporary solutions currently available it can be understood that although tremor suppression devices are currently available, these devices are having many shortcomings. Therefore, there is an apparent need for a low weight, less bulky orthosis for the suppression of hand tremors.

In developing such a low weight and less bulky device for hand tremor suppression, jamming mechanisms such as layer jamming, granular jamming, fiber jamming can be used. Such Jamming methods are driven by vacuum and show high stiffness variation. These mechanisms have become popular among many soft robotic applications due to their low weight and rapid response. Considering the bulkiness, ease of fabrication and stiffness variation, the layer jamming mechanism

shows significant results compared to other jamming mechanisms [20].

In this paper, we hypothesize that the layer jamming soft glove can suppress hand tremors. Furthermore, the development of a layer jamming-based soft glove for hand tremor suppression is presented in this paper, including the preliminary study on hand tremors, characterization of the layer jamming element and evaluation of the performance of the hand tremor suppression on hand tremor patients. Finally, the ergonomics of the layer jamming based soft glove and potential developments are discussed.

II. METHODS

This research was conducted to develop a tremor suppression soft wearable device. First, a tremor motion assessment was conducted to identify the behavior of the tremor motion [21]. Based on that, the stiffening method was chosen and the layer jamming mechanism was proposed as the stiffening method of the tremor suppression device [22]. To identify the most suitable layer combination for the element, an analytical model was proposed and validated using 8 types of layer combinations. A prototype tremor suppression glove was fabricated using the selected layer jamming element type and validated on a test rig for the identified frequency range based on the tremor motion assessment and the most effective layer jamming element placement (dorsal side/palmar side) was selected based on the results obtained from the test rig. Finally, the tremor suppression glove was tested with recruited patients at the Neurological Clinic of National Hospital, Sri Lanka.

A. Tremor Assessment

1) *Validation*: The tremor data was obtained from an Inertial Measurement Unit (IMU) based soft glove. The glove was initially validated using induced tremors by 3 healthy participants (See Fig.1). The IMU selected for data collection was MPU-6050 (TDK Corporation, Japan), a 6-DoF accelerometer and gyroscope. The data obtained using the IMUs was compared against a video tracking of a marker placed on the glove using Kinovea (Ver. 0.8.27). The video recording of the motion was obtained using a 240-fps smart phone camera. A Fast Fourier transformation (FFT) on Matlab (Ver. R2019a) of the raw signal of each method was compared and used for the validation. The participants were required to hold out their hands as seen in Fig.7(a) and were given instructions to simulate tremors as follows.

- 0 s - 5 s - hand at rest
- 5 s - 20 s - tremor motion with low frequency
- 20 s - 25 s - hand at rest
- 25 s - 40 s - tremor motion with high frequency
- 40 s - 45 s - hand at rest

2) *Experimental Protocol*: Upon validation, the glove was used in obtaining data from 15 hand tremor patients recruited from the neurological clinic of the National Hospital, Sri Lanka. The testing was conducted under the approval of ethical guidelines provided by the Ethical Review Committee of the National Hospital, Sri Lanka upon the

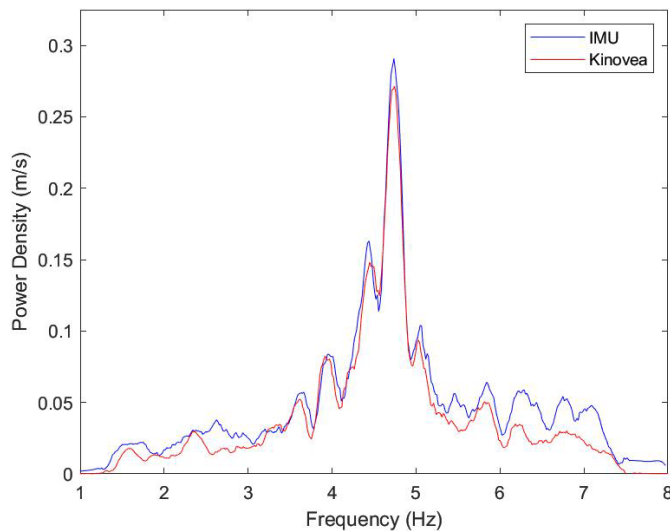


Fig. 1. Comparison in the frequency domain of acceleration in the Z axis for the validation.

TABLE I
INCLUSION AND EXCLUSION CRITERIA FOR THE HAND
TREMOR ASSESSMENT

Inclusion Criteria	Exclusion Criteria
Tremor is in right hand	Children (> 15 years of age)
Patients with Parkinson's Disease, Essential Tremor, Stroke or visible hand tremor	Patient with any chronic condition who has inability participate the study
	Refusal to participate the study

request by the authors for ethical clearance (ethical clearance ref. no. AAJ/ETH/COM/OCT/2019/07, date of approval - 24/10/2019). The profiles of the recruited patients are seen in Table II. Right hand dominant patients were recruited for this study. In Table II, the onset of tremor is given as the time since the beginning of tremor treatments. Patients were recruited to the tremor assessment according to the inclusion and exclusion criteria given in Table I.

The patients were requested to hold the right hand in the same posture as used in the validation process (see Fig. 7(a)) and data was recorded for 1 minute. Data was obtained for the index and middle fingers and patients were allowed to leave the study if they felt fatigue. Data collection was done using the IMU sensors attached to the glove. Participants were requested not to conduct any voluntary motion with their right hand during the recording process. Data was recorded from one sensor at a given instance for a time interval of one minute. The IMU sensors were connected to an Arduino Mega 2560 controller board and all the sensor readings were acquired using CoolTermWin software into a PC running Windows 10. The obtained data was analyzed in Matlab (Ver. R2019a) via FFT. Acceleration along the Z axis was further analyzed using FFT and the frequency with maximum power density was considered as the tremor frequency of each sensor as shown in Table III.

B. Layer Jamming Method

Stiffening methods using fabric gloves have been used to suppress hand tremors in previous studies [18]. The layer jamming mechanism was selected as the stiffening method

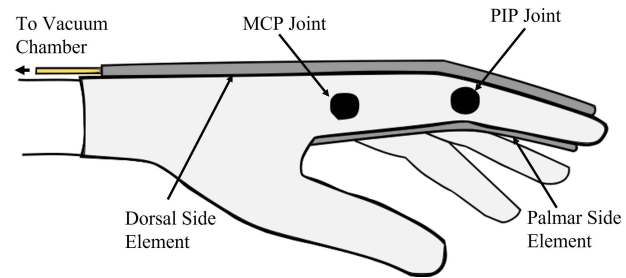


Fig. 2. Conceptual design of the element placement on the tremor suppression glove.

TABLE II
PROFILES OF THE RECRUITED PATIENTS FOR THE TREMOR
ANALYSIS (F_{ID} - FREQUENCY OF INDEX DISTAL,
 F_{IP} - FREQUENCY OF INDEX PROXIMAL)

Patient No.	Gender	Age	Onset of Tremor	F_{ID} (Hz)	F_{IP} (Hz)
1	M	24	7y	6.02	4.51
2	F	50	1y	5.55	-
3	F	64	3y	4.91	1.60
4	M	56	6m	4.10	3.13
5	M	56	4m	1.88	2.79
6	M	70	1y	1.89	-
7	M	78	1y	1.90	2.66
8	M	-	2y	4.41	6.32
9	F	73	2y	1.63	7.01
10	M	65	2y	2.68	2.32
11	M	-	5y	2.58	1.81
12	F	73	8m	7.33	5.69
13	M	58	1y	4.41	5.01
14	F	60	6m	3.23	2.39
15	F	56	17y	3.03	1.65

due to its inherent advantages of low weight and low bulkiness [23], [24]. Also, non-actuated layer jamming elements are flexible which will reduce the discomfort during voluntary motions. Therefore, layer jamming elements were integrated onto a soft fabric glove. These elements will create a resistant force that can absorb the energy of the involuntary movements and damp the tremor motion. The damping mechanism of this tremor suppression glove is given in Fig. 2. Hence, the apparent tremor in the vertical plane will be hidden and help improve the ADL of the patients. The layer jamming elements were proposed to be placed on either side of the finger (palmar or dorsal) and further testing was conducted to identify the most effective placement. Moreover, layer jamming elements are spanned across the metacarpophalangeal joint (MCP), proximal interphalangeal joint (PIP), distal interphalangeal joint (DIP) and palm as shown in Fig. 2. Additionally, since the design allows for open fingertips if necessary, the tactile sensation of the user is not impeded [25].

1) *Analytical Model*: Layer jamming mechanism activates with the negative pressure. It contains a stack of paper layers in a sealed polythene enclosure as shown in Fig. 3. The air gap between the layers will be removed upon applying negative pressure to this enclosure. Hence, the frictional forces between the adjacent layers will increase and create a resistance to the slippage between adjacent layers. Therefore, the layer jamming element will resist bending.

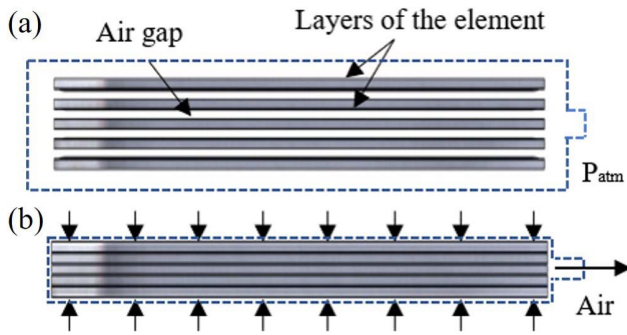


Fig. 3. Layer jamming element operation. (a) Before applying vacuum to the layer jamming element, (b) After applying vacuum to the layer jamming element.

An analytical model was developed for the selection of the suitable combination and the identification of the influence of the number of layers and the vacuum pressure applied for the layer jamming elements. Based on the analytical model, a suitable layer combination was selected. 8 layer combinations were used for this selection (See Table III). The model is based on the Euler-Bernoulli beam theory to describe the mechanical behavior of the layer jamming element after applying the negative pressure [26]. Vacuumed layer jamming elements reflect 3 phases of deformation: Pre slip phase, Transition phase, and Full slip phase. Hence, the rate of bending deflection varies with these phases.

In the pre slip phase, no slippage between adjacent layers is observed and all the layers stay cohesive. When the load is increasing, longitudinal shear stresses acting on the interface will increase. Layers begin to slip in certain regions in the transition phase when the longitudinal shear stress exceeds the maximum shear stress. The interface of the layers begins to slip along the full length of the element in the full slip phase.

The analytical model was derived for two uniformly distributed loads (self-weight of the element and applied load) as shown in Fig.4(a). For the pre slip phase, the transverse deflection is given by Eq.1.

$$\text{Transverse Deflection } (D_T) = \frac{L^4}{16EI} (F + w) \quad (1)$$

(E – Resultant Young's Modulus, I – Second Moment of Inertia, F – Applied Load, w – Self Weight, L – Longitudinal Length).

This analytical model was developed for multi-material and multi-layer combinations. Hence, the resultant Young's modulus and the second moment of inertia of the layer jamming element are given by Eq.2 and Eq.3 respectively. Proposed layer jamming elements are a combination of two types of materials which are denoted as material A and material B. Thus, E_A and E_B represent the Young's moduli of material A and B respectively. Thicknesses of layer material A and B are denoted by t_A and t_B respectively.

$$E = \frac{E_A t_A + E_B t_B}{t_A + t_B} \quad (2)$$

$$I = \frac{b(n_A t_A + n_B t_B)^3}{12} \quad (3)$$

TABLE III
COMBINATIONS OF LAYER JAMMING ELEMENTS TESTED

Element type	Mass (g)	Sandpapers (320 grade) abrasive side facing each other	Sandpapers (320 grade) abrasive side opposing each other	Sandpapers (120 grade) abrasive side facing each other	Sandpapers (120 grade) abrasive side opposing each other	Tracing papers (0.14 mm thickness)	Plastic films (0.15 mm thickness)
A	6.49	×10	-	-	-	-	-
B	6.33	-	×10	-	-	-	-
C	8.69	-	×5	-	-	×5	-
D	10.57	-	-	×10	-	-	-
E	10.57	-	-	-	×10	-	-
F	8.79	-	-	×5	-	×5	-
G	4.24	-	-	-	-	×5	×5
H	8.06	-	-	-	-	-	×10

2) *Experimental Verification Protocol*: Experimental validation was done for the verification of the analytical model. Based on this, a suitable layer combination was selected. According to the literature, the bending stiffness of elements should vary with the number of layers for the given analytical model [26]. Hence, for this validation, equal-sized elements (115mm × 15mm) was used and the number of layers were maintained as 10 as a preliminary value. The experimental setup used for this study is given in Fig.4(b). A wooden die was used to fold the elements in a 'C' shape with an overhanging length of 50 mm. The height of the C-shaped element was maintained as 22 mm through out the experiment. Standard weights were loaded onto the C-shaped structure from 0 g to 100 g with the 10 g intervals. Each loaded structure was captured using a digital camera (Canon EOS600D) to measure the deflections at the free end of this structure as shown in Fig.4(b). The deflection from the initial point was measured using the ImageJ software (Ver. 1.52a). The same experiment was repeated for 5 pressure values varying from 10 kPa (abs.) to 50 kPa (abs.). A single stage vacuum pump (FY-1C-N, VALUE) and a 25 l vacuum chamber were used to supply and maintain constant pressure throughout the experiment.

C. Tremor Suppression Glove and Fabrication

Layer jamming elements were fabricated with the dimensions of 115 mm × 20 mm. 10 layers were added for each element. The layer stack was wrapped with the polyethylene film which has a thickness of 0.05 mm and sealed using a thermal impulse sealer. Before the sealing, a small hole was created to attach the latex tubing (internal diameter 2 mm) to supply the vacuum. Super glue and silicon sealant were used for the sealing and fixing of the tube to the element. Each step of the fabrication process is given in Fig.5. The same procedure was followed to fabricate all the elements for each combination.

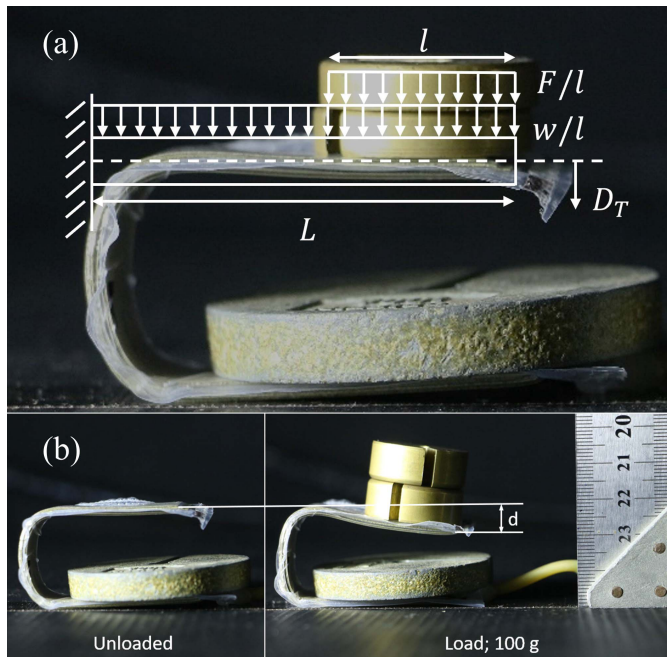


Fig. 4. (a) Free body diagram of the loading on the element in the experiment, (b) Experimental set up for the analytical model validation experiment.

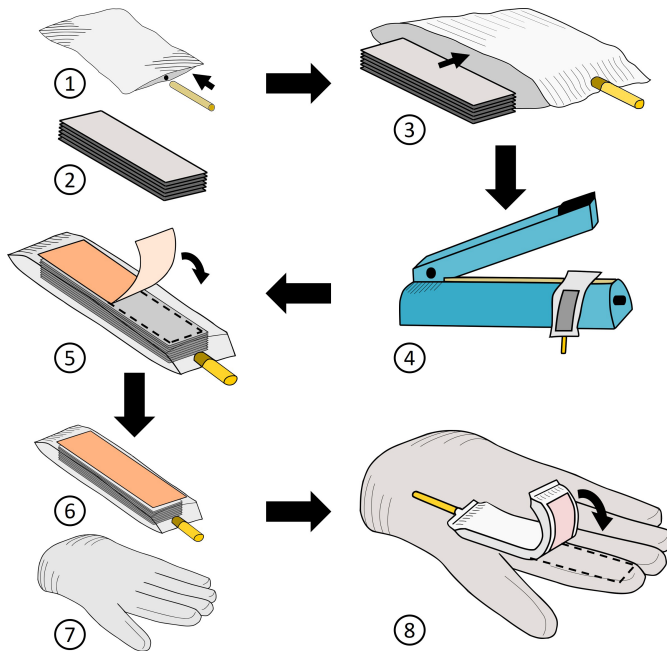


Fig. 5. Fabrication steps of the tremor suppression glove: (1) Attaching the latex tubing to the polythene covering, (2) Stack of layers, (3) Placing the Stack of layers inside partially sealed cover, (4) Sealing the polythene cover using an impulse sealer, (5) Attaching the adhesive strip to the polythene, (6) Fully sealing the Polythene cover with layer stack and attaching the adhesive strip, (7) Fabric glove, (8) Attaching the layer jamming element to the fabric glove on the dorsal side using silicon sealant.

In the soft glove fabrication, the layer jamming elements were attached onto a fabric glove. A piece of elastic adhesive bandage was resized into the size of the element and attached to the element as shown in Fig.8. Subsequently, the layer jamming element was attached to the soft glove by applying

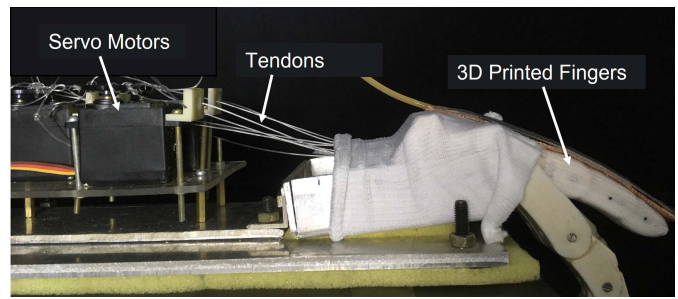


Fig. 6. Tendon-driven prosthetic hand as the test rig for the layer jamming placement evaluation.

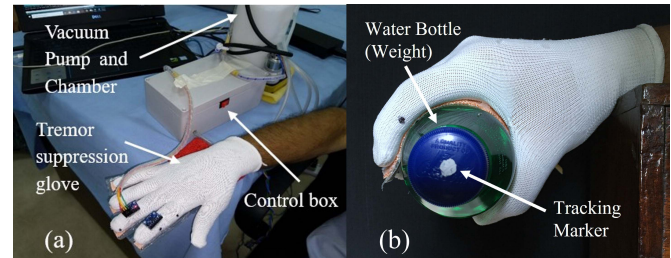


Fig. 7. Experimental setup for clinical trials. (a) For index and middle finger experiment, (b) For grasping experiment.

silicon sealant on the bandage and the fabric glove. The bandage helped to improve the adhesion and hold the element and the glove. Elements were attached to both the dorsal and palmar sides of the hand using this method.

1) Protocol for Selecting the Placement of Jamming Element:

The selection of the optimal placement of the layer jamming element, was determined using a test rig (See Fig.6). A prosthetic hand developed at the Bionics Laboratory, University of Moratuwa was used as the test rig for this experiment. The fingers of the selected prosthetic hand have 3 degrees of freedom for the flexion-extension of the MCP, PIP and DIP joints. The prosthetic hand is actuated using nylon tendons (diameter 0.7 mm) attached to servo motors (Tower Pro MG995) and controlled using an Arduino Mega 2560 micro-controller. This test rig only emulated the motion of the hand, disregarding the stiffness of the finger.

The prototype layer jamming glove was worn over the prosthetic hand and was actuated. Two configurations were tested with the element being attached either to the dorsal side or the palmar side respectively. The prosthetic hand was actuated at frequencies ranging from 3 Hz to 6 Hz based on the results obtained from the tremor data collection. Each experiment was conducted for 1 minute. A vacuum of 20 kPa (abs.) was supplied to the elements during the time interval of 15 s - 45 s. The tremor motion throughout the one-minute period was recorded by tracking a marker placed on the glove on Kinovea (Ver. 0.8.27) using a video recording from a smartphone camera (recorded at 240 fps). The experimental setup was similar to the validation experiment of the sensing glove. The reduction in the tremor amplitude was calculated as a percentage reduction and was used in selecting the optimal placement of the element. The ratio between mean amplitude reduction during the vacuum on period and the

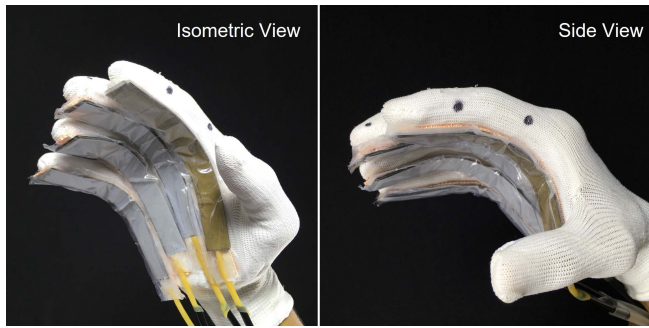


Fig. 8. Prototype glove used in clinical trials.

TABLE IV

PROFILES OF THE RECRUITED PATIENTS FOR CLINICAL TRIALS

Patient No.	Gender	Age	Onset of Tremor
1	F	65	6m
2	M	68	2y
3	M	65	5y
4	M	65	2y
5	M	69	4y
6	F	76	34y
7	M	43	5y
8	F	68	3y
9	M	69	7y
10	F	73	3y
11	F	68	1y

mean amplitude during the vacuum off period was obtained to calculate the percentage tremor amplitude reduction.

D. Clinical Trials

1) *Experimental Protocol*: Once the placement of the layer jamming element was determined, a final version of the glove was fabricated and used in the testing with tremor patients (See Fig.8). All experiments were conducted under conformance to ethical guidelines provided by the Ethical Review Committee of the National Hospital, Sri Lanka as mentioned previously. 11 patients with tremors on the right hand were recruited through the neurological clinic attached to Ward 84 of the National Hospital, Sri Lanka. The profiles of the recruited patients are summarized in Table IV.

The data gathering was two-fold. First, the patients were requested to hold the hand opened flat (palm downside) horizontally as used in the validation process (see Fig.7(a).) and data was recorded for 1 minute using a smartphone camera (recorded at 240 fps), with vacuum supplied to the elements during the period of 15 s – 45 s. Data was obtained for the index and middle fingers. As the second test, the patients were requested to hold a filled water bottle (500g) (See Fig.7(b)). Here too the glove was actuated during the period of 15 s - 45 s. The vertical (z-axis) acceleration of the tremor motion was acquired by tracking a marker in the recorded video using Kinovea (Ver. 0.8.27).

Data collection was conducted using the tracking of the marker on the glove and bottle. Participants were instructed not to conduct any voluntary motion with their right hand during the recording process. The data gathering process is similar to the one mentioned in the placement selection protocol mentioned previously. FFT analysis was conducted on the acquired

TABLE V

STATISTICAL ANALYSES CONDUCTED ON EXPERIMENTS

Experiment	Analysis
Preliminary study (15 Patients)	FFT analysis on raw acceleration data, Filters : Savitzky-Golay filter (order - 5, frame length - 31)
Performance evaluation on test rig	Mean amplitude analysis on raw acceleration data, Filters : Savitzky-Golay filter (order - 5, frame length - 21)
Clinical trials (11 Patients)	FFT analysis on raw acceleration data, t test analysis on peak frequencies for each task Filters : Savitzky-Golay filter (order - 5, frame length - 31)

vertical acceleration to obtain the percentage power reduction of the mean frequency of the tremor motion. For smoothing, the Savitzky-Golay filter was applied after converting the raw acceleration data into the frequency domain.

E. Statistical Analysis

All statistical analyses conducted on each experiment were included in Table V.

III. RESULTS

A. Preliminary Tremor Data Analysis

The analysis of the preliminary study on hand tremors from the cohort of 15 patients show that the mean of peak frequencies of the hand tremor is 5.05 ± 2.03 Hz. By gender, 4.62 ± 1.78 Hz and 5.69 ± 2.38 Hz are the mean of peak frequencies for male and female samples respectively. The dataset shows that the tremor frequencies vary between 1 - 8 Hz which is relatively lower than the global counterpart of 4 – 18 Hz.

B. Layer Jamming Selection

C, D, E, and F combinations were selected based on the weight of the element. Among the selected combinations, combinations C and F show higher stiffness compared to other combinations (See Fig.9(d)). Based on the analytical model, combination C has higher stiffness than combination F (See Fig.9(a)). Hence, combination C was selected for the tremor suppression glove considering the low weight and the higher stiffness. As shown in Fig. 10, unjammed state of the layer jamming element can not withstand the lowest weight (3.25 g) of the experiment and does not retain the shape without jamming.

For the validation of the analytical model, experimental data was compared with the results obtained from the model. All the experimental data were matched with the pre slip phase result as shown in Fig.9(b). Hence, a pressure variation between 10 kPa (abs.) and 50 kPa (abs.) does not affect the bending stiffness of the element. Moreover, similar to previous research, linear behavior can be observed in the experimental data for the tested pressure region (10 kPa (abs.) and 50 kPa (abs.)) and Fig.9(c)(d) show less variation in the deflection when varying the applied pressure [27]. Therefore, 20 kPa (abs.) is used for the further evaluation of the tremor suppression glove to generate the vacuum quickly and easily.

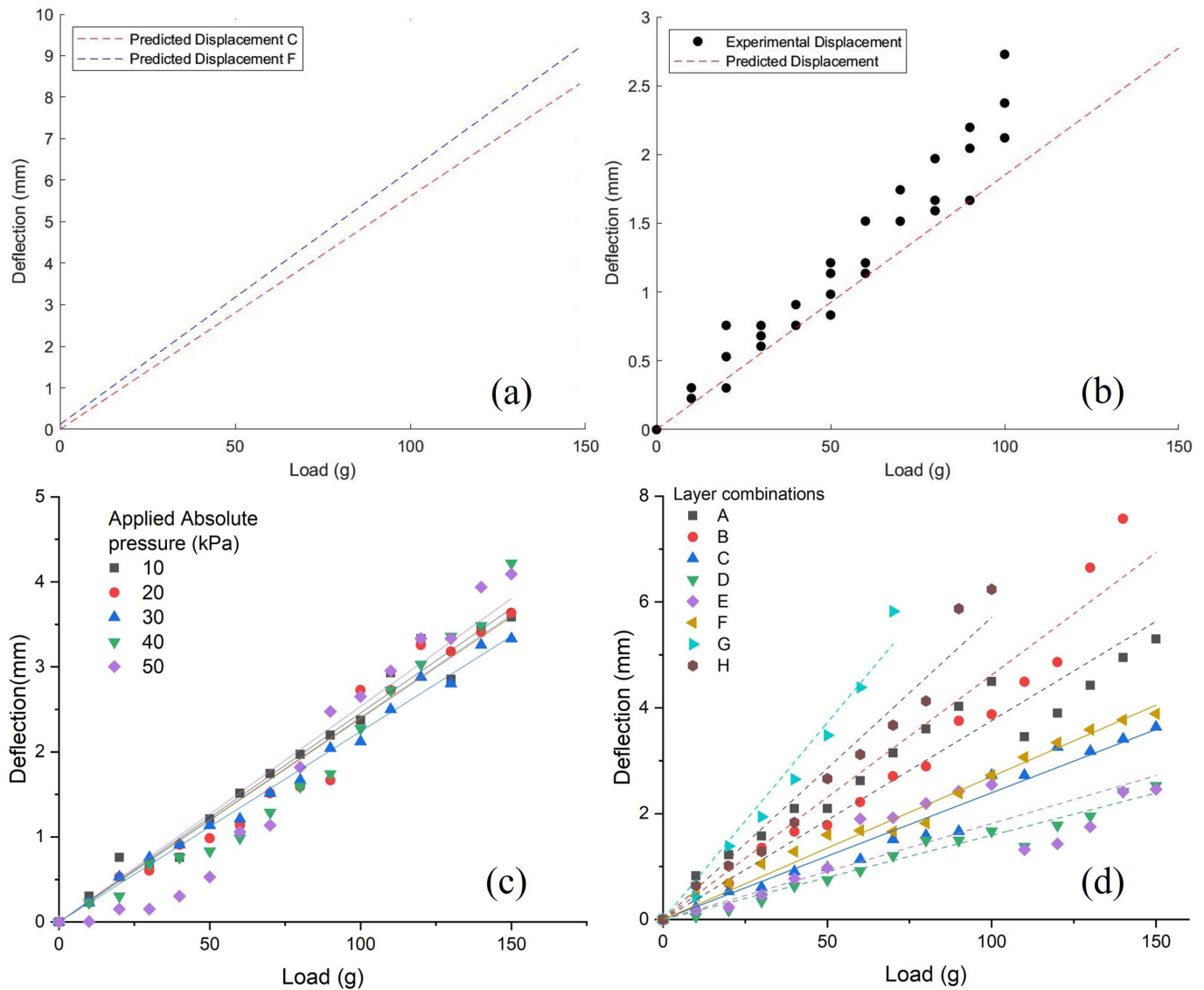


Fig. 9. Results of the layer jamming element selection. (a) Comparison between C and F using analytical model, (b) Validation of the Analytical model with experiment data for element C, (c) Measured deflection (mm) vs Applied load (g) for element C, (d) Measured deflection (mm) vs load (g) for all layer combination at 20 kPa (abs.).



Fig. 10. A model unjammed (inactive) layer jamming element with a 3.25g coin.

C. Selection of Layer Jamming Element Placement

Once combination C was selected from the previous experiment, it was used in fabricating a glove to determine the placement on the element. As the mean of peak frequencies obtained from the tremor data analysis was 5.05 ± 2.03 Hz, the test was conducted for a range

of 3 – 6 Hz. Results (see Fig.11) show that the percentage tremor amplitude suppression is considerably high in the palmar configuration over the dorsal configuration. A maximum tremor amplitude reduction of 27.41% and 78.32% was obtained in the dorsal and palmar configurations, respectively.

D. Clinical Trials of the Tremor Suppression Glove

The glove was fabricated with the layer jamming element on the palmar side and used for clinical trials. 11 patients were recruited as mentioned previously to study the performance of the hand tremor suppression glove. Here, the analysis of the FFT was conducted and the percentage power reduction of mean frequency was calculated. For further analysis on the behavior of hand tremors during the experiment, the one-minute time interval was divided in to two halves as 0 to 30 seconds and 30 to 60 seconds. The percentage reduction

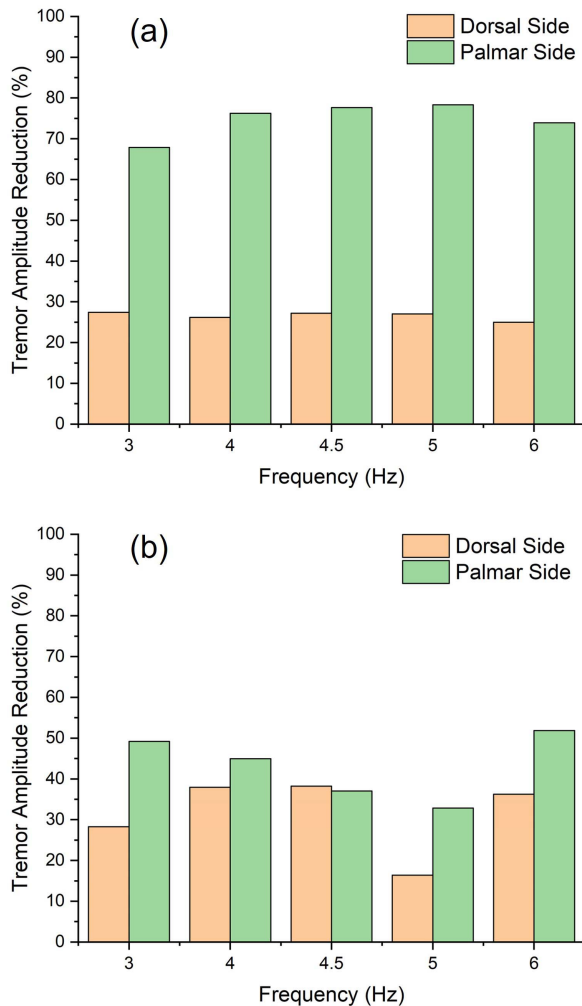


Fig. 11. Percentage tremor reduction on the test rig. (a) Combination C, (b) Combination F.

TABLE VI

POWER REDUCTION OF THE PEAK FREQUENCY FOR EACH PATIENT

Patient No.	Index (%)	Middle (%)	Grasping (%)
1	40.84	52.56	45.53
2	33.56	29.70	9.79
3	44.21	57.49	16.14
4	52.20	44.82	34.69
5	47.29	44.41	-
6	59.05	59.15	-
7	17.57	21.29	26.65
8	48.01	40.04	-
9	45.94	53.11	-
10	24.56	17.33	22.77
11	45.93	-	17.37
Mean (%)	41.74	41.99	24.71
Standard Deviation (%)	12.11	14.82	12.18
P-Value	0.0006	0.0007	0.067

of the maximum half in each patient is given in Table VI. The results show that a mean percentage power reduction of the mean frequency of $41.74 \pm 12.11\%$, $41.99 \pm 14.82\%$ and $24.71 \pm 12.18\%$ was recorded in the index and middle fingers in the initial placement of the grasping task, respectively. Almost all the p-values (t-distribution) for each experiment show significant results as shown in Table VI. The results

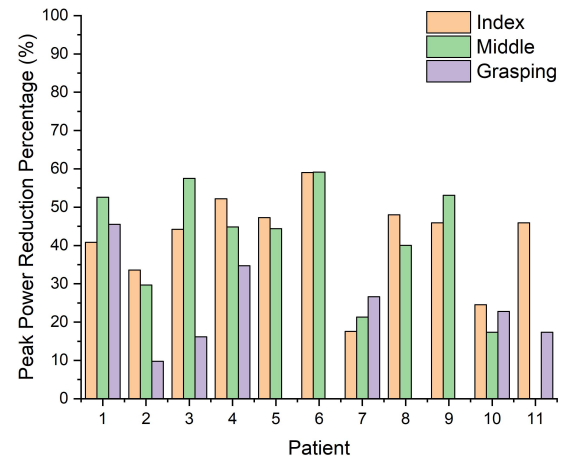


Fig. 12. Percentage tremor reduction for each patient.

show a clear reduction in the hand tremor of the patients under various conditions. Patients unable to grasp the bottle (due to fatigue or muscle weakness) were allowed to leave the study upon request. A maximum tremor power reduction of 59.15% was recorded in the study giving proof of the effectiveness of the system. The power reduction of the peak frequency is shown in Table VI and Fig.12 while Fig.14 shows the FFT for the trials with the maximum tremor power reduction for each task (Index, Middle, Grasping). Fig.13 depicts the distribution of the power reduction for the trials conducted.

Upon examining the peak frequencies of each patient, it can also be noted that there is no considerable change in the peak frequency of the hand tremor when the glove is in active mode.

IV. DISCUSSIONS

A. Analysis of Clinical Trials

The clinical results with the device on 11 patients show its capability to provide tremor suppression. The tests were conducted as three experiments (index, middle and grasping) and the power reduction of the mean frequency was measured. The results show a maximum tremor power reduction of 59.15%. This acts as the proof-of-concept for the device in its actions as a tremor suppression device and these results support the hypothesis that layer jamming soft glove can suppress hand tremors. The median power reduction in each testing protocol (index, middle and grasping) is approximately similar. The unavailable results are due to patients withdrawing from the study due to fatigue or being unable to perform the tasks due to physiological conditions such as muscle atrophy.

The results also show a high variation of tremor reduction percentage amongst the cohort of 11 patients. Factors such as the onset of tremors, age, and medical condition may have caused such effects and requires further studies for proof. For this study, only the tremor suppression of index and middle fingers were considered, assuming that the middle finger represents the functionality of the ring and little fingers. But further study will be required to analyze the tremor reduction in the remaining fingers as well. Although the degree of required stiffness level varies with the individual patient, the stiffness of the layer jamming mechanism was not tuned for

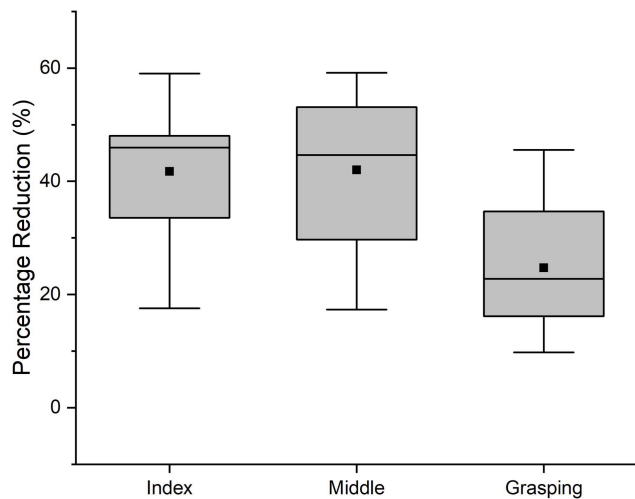


Fig. 13. Distribution of Power reduction of the peak frequency. Shaded squares indicate the mean percentage power reduction of the peak frequency of each task.

the individual patient. Hence, variations of the tremor motion among the study participants were not considered in this study.

B. Ergonomics of Device

A soft fabric glove was used for the design to ensure the physical comfort of the wearer. Considering the ergonomic comfort, it could be manufactured in different sizes to fit an individual's hand. All layer jamming elements could be customized according to the size of the hand, but it is required to consider the change in the stiffness when changes in width are made. It was designed to fit the size of the finger and avoid any disturbance when moving fingers separately [22]. Moreover, compliant materials were integrated with the layer jamming elements, and it does not restrict any voluntary movement and degree of freedom of the user's hand when in both active and inactive states. To quantitatively analyze the impedance on the dexterity of the hand, the fingertip displacements were compared with and without the glove when squeezing a half-cut rubber ball 20 times repeatedly to fatigue the hand as shown in Fig. 15. The average difference between the displacements of the 1st and 20th cycles of the experiments with and without the glove is 0.14 cm and 0.07 cm, respectively. Hence, the results of the fingertip displacement show that the glove does not considerably impede the fingertip displacement (See Table VII). In addition, the tremor suppression glove's low weight at 30 g with all the elements and low bulkiness increase the physical comfort and ease in conducting voluntary movement. The aforementioned tremor suppression devices have a higher weight, ranging above 850 g for exoskeletal devices and above 95 g for soft orthoses. Although, passive orthosis by Fromme *et al.* weighs only 33 g excluding the pneumatic components, the proposed tremors suppression glove is lighter than that. The air structure of the passive orthosis by Fromme *et al.* has a thickness of 24 mm during the active state while in the proposed device, the thickness of the layer jamming element is less than that which is 2 mm during the active state [18].

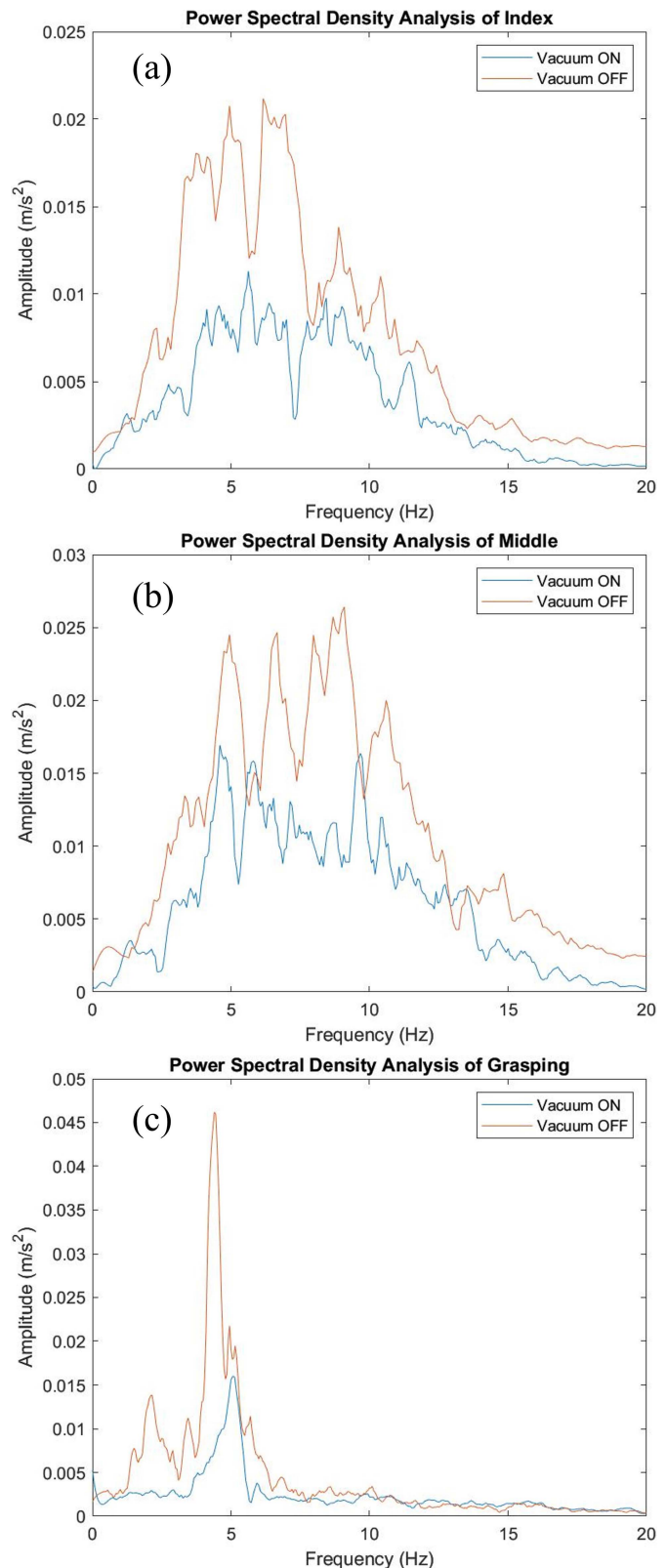


Fig. 14. FFT of the trial with maximum power reduction trial, (a) Index finger (Patient 6), (b) Middle finger (Patient 6), (c) Grasping task (Patient 1) (Blue: FFT of vacuum on state; Orange: FFT of vacuum off state).

Usability and reliability were considered when designing the glove as a human-centered device. Considering the usability, this device has a simple procedure to follow. First, the user

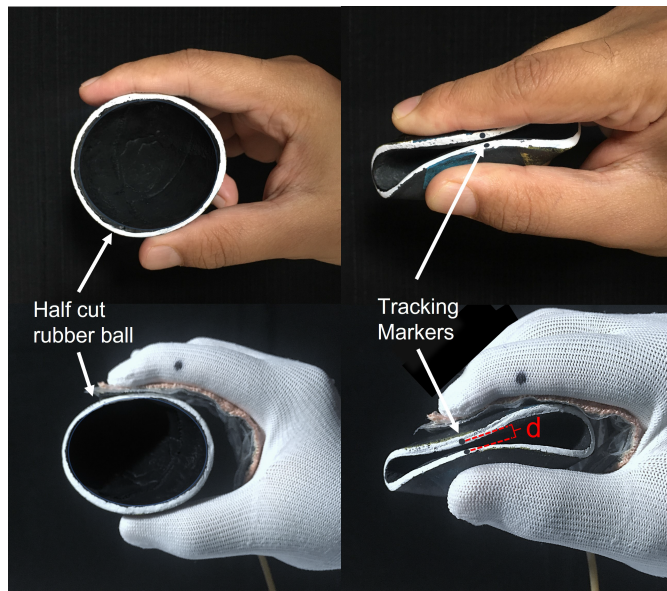


Fig. 15. Range of motion with and without the tremor suppression glove.

TABLE VII
FINGERTIP DISPLACEMENT ANALYSIS

Trial No.	Marker distance (d) with glove (cm)		Marker distance (d) without glove (cm)	
	1 st cycle	20 th cycle	1 st cycle	20 th cycle
1	0.04	0.2	0.01	0.11
2	0.05	0.19	0.05	0.12
3	0.05	0.18	0.02	0.06

must don the glove and supply vacuum for all the elements up to relevant pressure. After using the glove, the user can easily doff the glove as a regular glove after releasing the pressure. Easy donning and doffing of the glove were considered to support the elder community who comprise the majority of hand tremor patients. Evacuation of the elements can be done using a small electric vacuum pump (D2028B, AirpoTM). Within a second, all the elements can be evacuated or released from the vacuum. This device cannot cause any bodily harm to the user. Also, this device works as an active orthosis with the vacuum pump as the only major electrical component in this device. Hence, this system is reliable due to the minimum number of components. Even with leaks and damages, the device will function as it is vacuum-driven and not cause major injuries to the user. Hence, this device is safer and more reliable.

V. CONCLUSION

This paper presents the development of a layer jamming-based soft glove for tremor suppression. Material selection for layer jamming was done using an analytical model. An experiment was conducted to validate the analytical model using 8 types of layer combinations. Based on the results of the analytical model, combination C (Sandpaper – 320 grade + Tracing paper (0.14 mm thickness) (5 layers each, alternative)) and combination F (Sandpaper – 120 grade + Tracing paper (0.14 mm thickness) (5 layers each, alternative)) were selected for further analysis while combination C shows higher stiffness

than F. To verify the results, a tremor suppression glove was evaluated with both combinations C and F using a test rig. The tremor was simulated on a prosthetic hand with the tremor frequency range based on the mean of the peak frequency of 5.05 ± 2.03 Hz obtained from the tremor assessment. The performance evaluation also concluded that the palmar side element configuration has higher tremor amplitude reduction while verifying that combination C has a higher stiffness than F as expected. Test rig experiment shows a tremor amplitude reduction of 78.32% and 51.84% in the palmar side configuration for combinations C and F respectively. Moreover, the overall results of the test rig experiments validate the applicability of the proposed layer jamming mechanism for hand tremor suppression. Clinical trials were conducted to evaluate this tremor suppression glove with 11 patients at the Neurological Clinic of the National Hospital, Sri Lanka. Clinical trials show a tremor power reduction of $41.74 \pm 12.11\%$, $41.99 \pm 14.82\%$ and $24.71 \pm 12.18\%$ in the index, middle, and grasping task experiments, respectively. This performance evaluation with tremor patients confirms the validity of the proposed concept.

As future work, the ergonomics and the aesthetic appearance of the device could be improved to reduce discomfort with ADL on the patient. The proposed tremor suppression glove is planned to be developed as an untethered device using a low volume vacuum system to improve wearability. Furthermore, the suppression glove will be developed to control the stiffness based on the tremor severity, voluntary movement of the hand and the gripping patterns. The analytical model will be used to optimize the tremor suppression glove using force assessment.

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